



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to GURU RAGHAVENDRA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

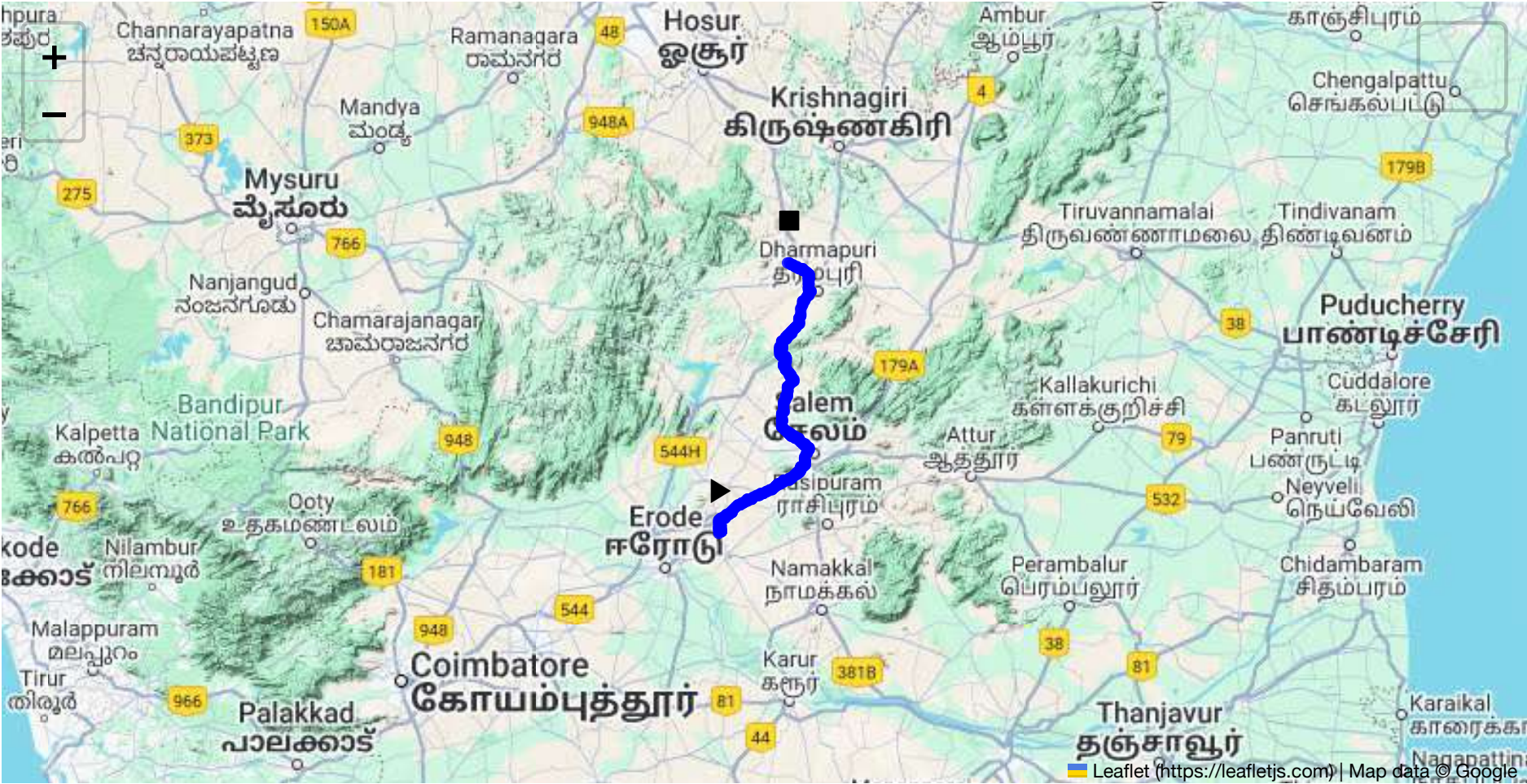
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 116.23 km
Estimated Duration: 2.3 hours
Adjusted Duration (Heavy Vehicle): 2.9 hours

Start: (11.4381, 77.8734)
End: (12.199469, 78.067903)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri, Tamil Nadu to Thattarapatti, Tamil Nadu spans approximately 116.23 kilometers. It usually takes about 2.29 hours for heavy vehicles. The journey passes through several rural and semi-urban areas, featuring a mix of state highways and major district roads.

Weather Conditions and Potential Hazards

Tamil Nadu experiences tropical weather, with high temperatures and humidity most of the year. **Typical weather conditions** include:

- **Summer (March to June):** High temperatures and strong winds, potentially impacting vehicle handling.
- **Monsoon (July to September):** Heavy rains can lead to flooding and reduced visibility, and slippery roads.
- **Winter (October to February):** Mild and pleasant, but early morning fog can reduce visibility.

Weather-related hazards to consider include waterlogged roads during monsoons and heat-related vehicle strain in summer.

Traffic Patterns

Traffic tends to peak during:

- **Morning (8 AM - 10 AM):** Increased local and long-distance commuter traffic.
- **Evening (5 PM - 8 PM):** High return and commercial traffic.

Congestion-prone areas include:

- Townships and intersections close to market zones and near the highways.

Road Quality and Infrastructure

The roads along this route are a mix of well-maintained highways and lesser-maintained district roads:

- **Highways:** Generally in good condition, marked lanes, and proper signage.
- **District Roads:** Potentially narrow with few passing opportunities, limited shoulder width, and occasional potholes.

Alternative Routes for Emergencies

In case of road closures or emergencies:

- Consider using alternate routes via nearby villages or other state highways, depending on the exact road closures.
- Local roads can often be narrower and less maintained, necessitating caution.

Local Regulations Affecting Hazardous Material Transport

- **Permits** are required for transporting hazardous materials.
- **Restricted areas** may include zones near residential areas or schools.
- **Driving hours** limitations might impact travel during restricted times, especially concerning night transport.

Historical Incidents

There have been incidents involving:

- **Heavy vehicles losing control during monsoon conditions.**
- **Rollover incidents** due primarily to narrow roads and sharp turns.

Environmental Considerations and Sensitive Areas

Be cautious near:

- **Agricultural fields** where dust and debris may impact travel.
- **Water bodies** where runoff or spills could have environmental impacts.

Communication Coverage

- **Urban Areas:** Generally reliable mobile and data coverage.
- **Rural Sections:** Possible dead zones due to sparse cell towers. Communication may be limited, affecting GPS and emergency communication.

Estimated Emergency Response Times

- **Urban Areas:** 20-30 minutes due to better infrastructure closer to townships.
- **Rural Areas:** 45-60 minutes, as remoteness slows access for emergency services.

Overall Summary of Risk Assessment

The route is relatively manageable but presents some challenges typical of rural Indian roads. The primary risks include adverse weather, road quality in rural sections, and communication challenges in remote areas. However, with proper preparation, adherence to traffic regulations, and proactive emergency planning, these risks can be effectively managed to ensure safety for drivers and others on the route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	Medium	11.45000, 77.87213	30 KM/Hr
4	Turn	High	11.45028, 77.87202	15 KM/Hr
5	Turn	Medium	11.45309, 77.87325	30 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	Medium	11.46765, 77.87037	30 KM/Hr
8	Turn	High	11.47403, 77.86841	15 KM/Hr
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.53512, 77.95717	30 KM/Hr
12	Turn	Medium	11.53606, 77.95869	30 KM/Hr
13	Turn	Medium	11.64183, 78.11810	30 KM/Hr
14	Turn	Medium	11.64338, 78.11998	30 KM/Hr
15	Turn	Medium	11.92556, 78.06499	30 KM/Hr
16	Turn	Medium	11.95854, 78.04990	30 KM/Hr
17	Turn	Medium	11.96015, 78.04945	30 KM/Hr
18	Turn	Medium	11.96370, 78.05142	30 KM/Hr
19	Turn	Medium	11.96433, 78.05113	30 KM/Hr
20	Turn	High	11.96542, 78.04938	15 KM/Hr
21	Turn	High	11.96650, 78.04927	15 KM/Hr
22	Turn	Medium	11.96804, 78.05080	30 KM/Hr
23	Turn	Medium	12.12036, 78.13284	30 KM/Hr
24	Turn	Medium	12.15019, 78.12980	30 KM/Hr
25	Turn	Medium	12.16888, 78.12925	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
5	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
6	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
7	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
9	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
10	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Global Medical Centre Jagirammalayam	11.6894409, 78.1108847	30 km/h	Medium
30	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
33	pharmacy	MJ Medicals	11.7232728, 78.0698965	30 km/h	Medium
34	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
36	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
37	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
39	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
41	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
42	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
43	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	college	Govt Engineering College	11.7128537, 78.0877084	30 km/h	Medium
32	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
35	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
38	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
40	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
44	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium

Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45000, 77.87213



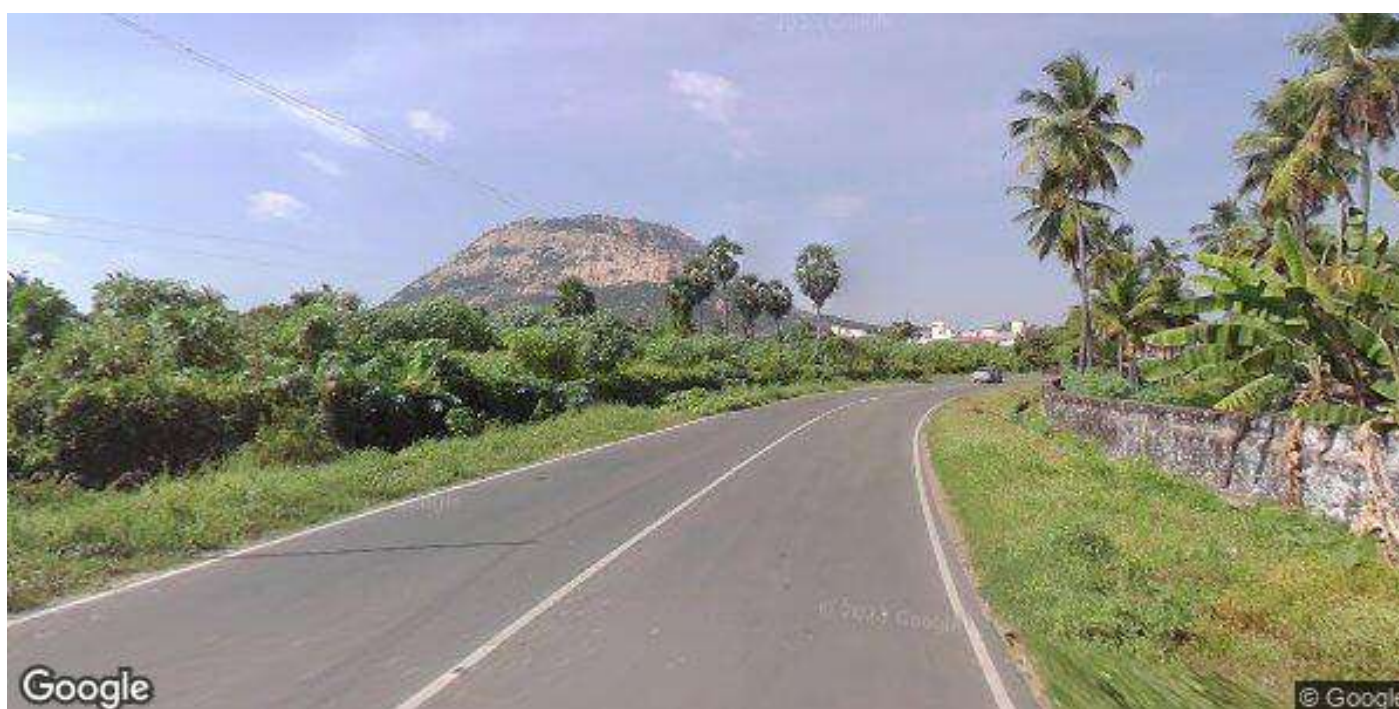
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53512, 77.95717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53606, 77.95869



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64338, 78.11998



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92556, 78.06499



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12036, 78.13284



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.15019, 78.12980



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.16888, 78.12925

